Econ 814 Recitation 6 Notes

Frank Gentile

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1 Midterm 22 Q2

[1]Consider an overlapping generations model with production and population growth (the size of cohort t is n percent higher than the size of cohort $t-1$). The endowments are given in units of labor: each agent is endowed with one unit of time when young and one unit of time when old. In addition, the initial old are endowed with K_1 units of capital. The production function $F(K_t, L_t)$ is

$$F(K_t, L_t) = K_t^{\alpha} L_t^{1-\alpha}$$

There is no asset in the economy other than capital, which accumulates according to $K_{t+1} = I_t$(we assume that there is full depreciation). The young at time t buy the capital stock (K_{t+1}) and then rent it at time $t+1$ to competitive firms at the rental rate r_{t+1}. Agents preferences over consumption streams are: $u(c_y, c_o) = u(c_{yt}) + \beta u(c_{ot+1})$

1.1 Part a

Define a competitive equilibrium for this economy.

An equilibrium is a sequence of prices $\{r_t, w_t\}_{t=0}^{\infty}$ and allocations $\{c_{1t}, c_{2t}, s_t, K_t, L_t\}_{t=0}^{\infty}$ such that:
Taking prices as given, the allocations s_t, c_{1t}, and c_{2t+1} solve the consumer's problem for all $t \geq 0$.

$$\begin{aligned} & \max_{c_{1t}, c_{2t+1}, s_t} u(c_{yt}) + \beta u(c_{ot+1}) \\ & s.t. \\ w_t &= c_{1t} + s_t \\ c_{2t+1} &= r_{t+1} s_t + w_{t+1} \end{aligned}$$

[1]Thanks to Professor Jorge Soares for problems and solutions

Taking prices as given, the allocations K_t and L_t solve the firm's profit maximization problem.

$$\begin{aligned} r_t &= \alpha K_t^{\alpha-1} L_t^{1-\alpha} \text{ as } \delta = 1 \\ w_t &= (1-\alpha) K_t^{\alpha} L_t^{-\alpha} \end{aligned}$$

Markets clear, i.e., for all t, we have
Labor market: $L_t = N_t + N_{t-1} = N_{t-1}(2+n)$ since young and old agents provide 1 unit of labor each
Goods market: $K_t = s_t N_{t-1}$

1.2 Part b

Assume from now on that $u(c) = \ln c$. Derive the young's demand for capital as a function of the rental rate and the wages.
The consumer's problem can be written as :

$$\max_{\{s_t\}} u(w_t - s_t) + \beta u(r_{t+1} s_t + w_{t+1})$$

taking the f.o.c. w.r.t. s_t:

$$u_c(w_t - s_t) = \beta u_c(r_{t+1} s_t + w_{t+1}) r_{t+1}$$

For our utility function we get:

$$\frac{1}{w_t - s_t} = \frac{\beta r_{t+1}}{r_{t+1} s_t + w_{t+1}}$$

$\Leftrightarrow$

$$r_{t+1} s_t + w_{t+1} = \beta r_{t+1}(w_t - s_t)$$

$\Leftrightarrow$

$$s_t = \frac{\beta}{1+\beta} w_t - \frac{1}{1+\beta} \frac{w_{t+1}}{r_{t+1}}$$

1.3 Part c

Write down an equation that determines aggregate capital in $t+1$, k_{t+1}, as a function of current capital, k_t, and other parameters.
In equilibrium we have

$$\begin{aligned} K_{t+1} &= S_t \text{ where } S_t = s_t N_t \\ L_t &= N_t + N_{t-1} = N_{t-1}(2+n) \end{aligned}$$

since young and old agents provide 1 unit of labor each.
So in per worker (also per capita in this case) terms we have

$$k_{t+1} = \frac{K_{t+1}}{L_{t+1}} = \frac{S_t}{L_{t+1}} = \frac{s_t N_t}{N_t\left(2+n\right)} = \frac{s_t}{2+n}$$

$$\begin{aligned} r_t &= \alpha K_t^{\alpha-1} L_t^{1-\alpha} = \alpha k_t^{\alpha-1} \\ w_t &= \left(1-\alpha\right) K_t^{\alpha} L_t^{-\alpha} = \left(1-\alpha\right) k_t^{\alpha} \end{aligned}$$

So

$$(2+n)k_{t+1} = \frac{\beta}{1+\beta} w_t - \frac{1}{1+\beta}\frac{w_{t+1}}{r_{t+1}}$$

$\Leftrightarrow$

$$\begin{aligned} (2+n)k_{t+1} &= \frac{\beta}{1+\beta}(1-\alpha)k_t^{\alpha} - \frac{1}{1+\beta}\frac{(1-\alpha)k_{t+1}^{\alpha}}{\alpha k_{t+1}^{\alpha-1}} \\ &= \frac{\beta}{1+\beta}\left(1-\alpha\right)k_t^{\alpha} - \frac{1}{1+\beta}\frac{1-\alpha}{\alpha}k_{t+1} \end{aligned}$$

$\Leftrightarrow$

$$\left[(2+n) + \frac{1}{(1+\beta)}\frac{1-\alpha}{\alpha}\right] k_{t+1} = \frac{\beta}{1+\beta}\left(1-\alpha\right)k_t^{\alpha}$$

$\Leftrightarrow$

$$k_{t+1} = \frac{\beta\left(1-\alpha\right)\alpha}{\left(1+\beta\right)\left(2+n\right)\alpha + 1 - \alpha} k_t^{\alpha}$$

1.4 Part d

Now, write down an equation determining capital in steady state, k_{ss}, and solve for k_{ss} in terms of primitives
In steady-state $k_{t+1} = k_{ss}$ so

$$k_{ss} = \frac{\beta\left(1-\alpha\right)\alpha}{\left(1+\beta\right)\left(2+n\right)\alpha + 1 - \alpha} k_{ss}^{\alpha}$$

$\Leftrightarrow$

$$k_{ss}^{1-\alpha} = \frac{\beta\left(1-\alpha\right)\alpha}{\left(1+\beta\right)\left(2+n\right)\alpha + 1 - \alpha}$$

$\Leftrightarrow$

$$k_{ss} = \left[\frac{\beta\left(1-\alpha\right)\alpha}{\left(1+\beta\right)\left(2+n\right)\alpha + 1 - \alpha}\right]^{\frac{1}{1-\alpha}}$$

1.5 Part e

Under which conditions will this economy be dynamically inefficient?
The economy is dynamically inefficient if

$$k_{ss} > k_{GR}$$

where

$$k_{GR} = \arg\max \left[c = f(k) - (n+\delta)k\right]$$

That is

$$k_{GR} : f'(k_{GR}) = \delta + n$$

So the economy is dynamically inefficient if

$$r = f'(k_{ss}) < \delta + n$$

that is

$$\alpha k_{ss}^{\alpha-1} = \frac{(1+\beta)(2+n)\alpha + 1 - \alpha}{\beta(1-\alpha)} < 1 + n$$

1.6 Part f

There is a government that levies a lump-sum tax τ_t on each young consumer in period t, and uses the proceeds to pay a lump-sum transfer TR_t to each old consumer in period t. The government does not borrow or lend.

Write down the governments budget constraint.
The government's budget constraint is

$$N_t \tau_t = N_{t-1} TRt,$$

or

$$(1+n)\,\tau_t = TR_t$$

1.7 Part g

Follow the same steps as in a),b) and c) and write down an equation that determines aggregate capital in $t+1$, k_{t+1}, as a function of current capital, k_t, a government policy $\{\tau_t\}_{t=0}^{\infty}$ that satisfies the government budget constraint in each period t, and other parameters.
The consumer's problem can be written as :

$$\max_{s_t} u(w_t - s_t - \tau_t) + \beta u(r_{t+1} s_t + w_{t+1} + TR_{t+1})$$

taking the f.o.c. w.r.t. s_t:

$$u_c(w_t - s_t - \tau_t) = \beta u_c(r_{t+1}s_t + w_{t+1} + TR_{t+1})r_{t+1}$$

For our utility function we get:

$$s_t = \frac{\beta}{1+\beta}(w_t - \tau_t) - \frac{1}{r_{t+1}(1+\beta)}(TR_{t+1} + w_{t+1})$$

A balanced government budget constraint implies:

$$s_t = \frac{\beta}{1+\beta}(w_t - \tau_t) - \frac{1}{r_{t+1}(1+\beta)}((1+n)\tau_{t+1} + w_{t+1})$$

In equilibrium we have

$$(2+n)k_{t+1} = s_t$$

and

$$\begin{aligned} r_t &= \alpha k_t^{\alpha-1} \\ w_t &= (1-\alpha)k_t^{\alpha} \end{aligned}$$

So

$$(2+n)k_{t+1} = \frac{\beta}{1+\beta}((1-\alpha)k_t^{\alpha} - \tau_t) - \frac{1}{\alpha k_{t+1}^{\alpha-1}(1+\beta)}\left((1+n)\tau_{t+1} + (1-\alpha)k_{t+1}^{\alpha}\right)$$

1.8 Part h

Now, suppose that the government sets $\tau_t = \tau$ for all t and $TR_t = TR$ for all t, and show that τ and TR can be set so as to achieve the golden rule steady state, as a competitive equilibrium. Derive a set of equations that solves for the corresponding levels of k_{ss}, τ and TR and solve those equations. What happens for levels of the lump-sum tax higher than the ones you found?

The golden rule steady-state level of capital is such that:

$$f'(k_{GR}) = 1 + n,$$

$\Leftrightarrow$

$$\alpha k_{GR}^{\alpha-1} = 1 + n$$

In order to achieve this steady-state in a competitive equilibrium we need k_{SS} to solve

$$\begin{aligned} \alpha k_{SS}^{\alpha-1} &= 1+n \\ (2+n)k_{SS} &= \frac{\beta}{1+\beta}((1-\alpha)k_{SS}^{\alpha} - \tau) - \frac{1}{\alpha k_{SS}^{\alpha-1}(1+\beta)}((1+n)\tau + (1-\alpha)k_{SS}^{\alpha}) \end{aligned}$$

$\Leftrightarrow$

$$\begin{aligned} k_{SS} &= \left(\frac{1+n}{\alpha}\right)^{\frac{1}{\alpha-1}} \\ (2+n)k_{SS} &= \left[\frac{\beta}{1+\beta} - \frac{1}{\alpha k_{SS}^{\alpha-1}(1+\beta)}\right](1-\alpha)k_{SS}^{\alpha} - \left[\frac{\beta}{1+\beta} + \frac{1+n}{\alpha k_{SS}^{\alpha-1}(1+\beta)}\right]\tau \end{aligned}$$

As $\alpha k_{SS}^{\alpha-1} = 1+n$ we have

$$\begin{aligned} k_{SS} &= \left(\frac{1+n}{\alpha}\right)^{\frac{1}{\alpha-1}} \\ (2+n)k_{SS} &= \left[\frac{\beta}{1+\beta} - \frac{1}{\alpha k_{SS}^{\alpha-1}(1+\beta)}\right](1-\alpha)k_{SS}^{\alpha} - \tau \end{aligned}$$

$\Leftrightarrow$

$$\begin{aligned} k_{GR} &= \left(\frac{1+n}{\alpha}\right)^{\frac{1}{\alpha-1}} \\ \tau_{GR} &= (2+n)k_{GR} - \left[\frac{\beta}{1+\beta} - \frac{1}{\alpha k_{GR}^{\alpha-1}(1+\beta)}\right](1-\alpha)k_{GR}^{\alpha} \end{aligned}$$

To see what happens to when $\tau > \tau_{GR}$,we need to see how k_{SS} responds to a change in tax. Intuitively, I was expecting an increase in the transfer from the young to the old to decrease the incentives to save, and therefore to bring steady-state capital down. This would be shown by $\frac{\partial k_{SS}}{\partial \tau} < 0$. However this derivative is not trivial, and does not lead to an unambiguous response of capital to a change in the transfer.

$$(2+n)k_{SS} = \frac{\beta}{1+\beta}((1-\alpha)k_{SS}^{\alpha} - \tau) - \frac{1}{\alpha k_{SS}^{\alpha-1}(1+\beta)}\left((1+n)\tau + (1-\alpha)k_{SS}^{\alpha}\right)$$

$$\Leftrightarrow$$

$$(1+\beta)(2+n)k_{SS} = \beta((1-\alpha)k_{SS}^{\alpha} - \tau) - \frac{1}{\alpha k_{SS}^{\alpha-1}}(1+n)\tau - \frac{1-\alpha}{\alpha}k_{SS}$$

$$\Leftrightarrow$$

$$\beta(1-\alpha)k_{SS}^{\alpha} - \left[(1+\beta)(2+n) + \frac{1-\alpha}{\alpha}\right]k_{SS} = \left[\beta + \frac{1}{\alpha k_{SS}^{\alpha-1}}(1+n)\right]\tau$$

$$\begin{aligned} \tau &> \tau_{GR} \\ &\Leftrightarrow \\ (1+\beta)(2+n)k_{SS} - \left[\beta - \frac{1}{\alpha k_{SS}^{\alpha-1}}\right](1-\alpha)k_{SS}^{\alpha} &> (1+\beta)(2+n)k_{GR} - \left[\beta - \frac{1}{\alpha k_{GR}^{\alpha-1}}\right](1-\alpha)k_{GR}^{\alpha} \\ &\Leftrightarrow \\ (1+\beta)(2+n)k_{SS} - \left[\beta(1-\alpha)k_{SS}^{\alpha} - \frac{1-\alpha}{\alpha}k_{SS}\right] &> (1+\beta)(2+n)k_{GR} - \left[\beta(1-\alpha)k_{GR}^{\alpha} - \frac{1-\alpha}{\alpha}k_{GR}\right] \\ &\Leftrightarrow \\ \left[(1+\beta)(2+n) + \frac{1-\alpha}{\alpha}\right]k_{SS} - \beta(1-\alpha)k_{SS}^{\alpha} &> \left[(1+\beta)(2+n) + \frac{1-\alpha}{\alpha}\right]k_{GR} - \beta(1-\alpha)k_{GR}^{\alpha} \end{aligned}$$

2 Final 22 Q2

(30 points) Consider the following overlapping generations model. In period t there are N_t two-period-lived consumers born, where $N_t = (1+n)^t N_0$, with N_0 given and $n > 0$. At $t = 0$ there are some one-period-lived old agents who are collectively endowed with K_0 units of capital, and who maximize consumption at $t = 0$. A consumer born in period t has preferences given by

$$u(c_{yt}, c_{ot+1}) = \ln c_{yt} + \ln c_{ot+1}$$

where c_{yt} (c_{ot}) denotes consumption of a young (old) consumer in period t. Each consumer has one unit of time available when young that is allocated to work. The production technology is given by

$$Y_t = K_t^{\alpha} N_t^{1-\alpha},$$

where Y_t is output, K_t is the capital input, N_t is the labor input, and $0 < \alpha < 1$. One unit of the consumption good can be converted into one unit of capital, and vice-versa. Capital fully depreciates after one period. Let w_t denote the real wage rate and r_t the capital rental rate.

2.1 Part a

(5 points) Determine the capital/labor ratio, the consumption of the young, and the consumption of the old in a socially optimal steady state. Assume that the social planner chooses a steady-state that maximizes $(\log c_y + \log c_o)$.
The resource constraint faced by the social planner is

$$N_t c_{y,t} + N_{t-1} c_{o,t} = (N_t k_t)^{\alpha} N_t^{1-\alpha} + (1-\delta) N_t k_t - N_{t+1} k_{t+1}$$

which, when divided by N_t, becomes:

$$\begin{aligned} c_{y,t} + \frac{c_{o,t}}{1+n} &= k_t^{\alpha} + (1-\delta) k_t - (1+n) k_{t+1} \\ &= k_t^{\alpha} - (1+n) k_{t+1} \end{aligned}$$

So, in steady state, the resource constraint faced by the social planner is

$$c_y + \frac{c_o}{1+n} = k^{\alpha} - (1+n)k;$$

where c_y is consumption by the young, c_o is consumption by the old, and k is the capital/labor ratio. The socially optimal steady state allocation is obtained as the solution to the problem of solving

$$\begin{aligned} &\max_{c_y, c_o, k} (\ln c_y + \ln c_o) \\ &\textit{subject to } \; c_y + \frac{c_o}{1+n} = k^{\alpha} - (1+n)k. \end{aligned}$$

which is equivalent to

$$\max_{c_o,k}\left[\ln\left(k^\alpha-(1+n)k-\frac{c_o}{1+n}\right)+\ln c_o\right]$$

The FOCs are

$$\begin{aligned} k &: \alpha k^{\alpha-1}-(1+n)=0 \\ c_o &: \frac{-\frac{1}{1+n}}{c_y}+\frac{1}{c_o}=0 \end{aligned}$$

The optimal capital/labor ratio is given by the solution to

$$\alpha k^{\alpha-1}=(1+n),$$

or

$$k=\left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}},$$

note that this is the Golden Rule level of capital, which makes sense as given the objective function the social planner would want to maximize steady-state consumption and then distribute it optimally across old and young. If we define

$$\phi=\left(\frac{\alpha}{1+n}\right)^{\frac{\alpha}{1-\alpha}}(1-\alpha),$$

we can use the optimality condition

$$c_o=c_y(1+n)$$

and the social planner's budget constraint to compute the optimal consumption of the young and the old:

$$c_y=\frac{\phi}{2},\ c_o=\frac{(1+n)\phi}{2}$$

2.2 Part b

(5 points) In period t, the government levies a lump-sum tax of τ_{yt} on each young person, and a lump-sum tax of τ_{ot} on each old person. Write down the government's budget constraint.
The government's budget constraint is given by

$$0=N_t\tau_{yt}+N_{t-1}\tau_{ot}. \quad (2)$$

There is no need to make any assumptions about the sign of the lump-sum taxes at this point.

2.3 Part c

(5 points) Determine the optimal savings of a young consumer given market prices and the taxes described above.
A young consumer solves[2]

$$\max_{s_t}\{\ln(w_t - s_t - \tau_{yt}) + \ln[(1 + r_{t+1})\, s_t - \tau_{ot+1}]\}$$

The FOC condition is

$$-\frac{1}{w_t - s_t - \tau_{yt}} + \frac{(1 + r_{t+1})}{(1 + r_{t+1})\, s_t - \tau_{ot+1}} = 0$$

and the solution is

$$s_t = \frac{w_t - \tau_{yt}}{2} + \frac{\tau_{ot+1}}{2\,(1 + r_{t+1})} \quad (3)$$

2.4 Part d

(15 points) Suppose that the government introduces a pay-as-you-go social security system whereby transfers are made between the young and the old in each period, so that $\tau_{ot} = -(1 + n)\tau_{yt}$; $\tau_{yt} = \tau$ for all t.

1. **(5 points) Show that the government budget constraint is satisfied in each period.**
   To verify that the government budget constraint is met, substitute

   $$\begin{aligned} \tau_{ot} &= -(1 + n)\tau, \\ \tau_{yt} &= \tau \end{aligned}$$

   in

   $$0 = N_t\tau_{yt} + N_{t-1}\tau_{ot}$$

   $$N_t\tau_{yt} + N_{t-1}\tau_{ot} = N_t\tau - N_{t-1}(1 + n)\tau = N_t\tau - N_t\tau = 0 \text{ QED} \quad (1)$$

2. **(5 points) Determine whether or not the government can set τ so as to achieve the socially optimal steady state in a competitive equilibrium. If the answer is yes, determine the optimal level of τ and whether the optimal τ is positive or negative.**
   In equilibrium, we have

   $$k_{t+1}(1 + n) = s_t, \quad (4)$$

   and from the firm's profit maximization problem,

   $$\begin{aligned} w_t &= (1 - \alpha)k_t^{\alpha}, \quad (5) \\ r_t &= \alpha k_t^{\alpha-1} - \delta \\ &= \alpha k_t^{\alpha-1} - 1. \quad (6) \end{aligned}$$

[2]Note here I think it should be r_{t+1} rather than $1+r_{t+1}$ below due to complete depreciation

Notice that we exclude the depreciation rate of capital here. We need to take into account the depreciation of capital so either we introduce it here or in the consumers and government budget constraints:

$$\begin{aligned} c_{o,t+1} &= (1 + r_{t+1} - \delta)\, s_t - \tau_{ot+1} \\ 0 &= N_t \tau_{yt} + N_{t-1} \tau_{ot}. \end{aligned}$$

Substituting using (2), (5) and (6) in (4), we get

$$k_{t+1}(1+n) = \frac{(1-\alpha)k_t^{\alpha}}{2} - \frac{\tau}{2} - \frac{\tau(1+n)}{2\alpha k_{t+1}^{\alpha-1}} \quad (7)$$

In the optimal steady state, we have

$$k_t = k_{t+1} = k = \left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}},$$

and so we can substitute in (7) and solve for the tax $\tau = \tau^*$ that yields this as a competitive equilibrium. We get

$$\left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}}(1+n) = \frac{(1-\alpha)\left(\frac{\alpha}{1+n}\right)^{\frac{\alpha}{1-\alpha}}}{2} - \frac{\tau}{2} - \frac{\tau(1+n)}{2\alpha\left(\frac{\alpha}{1+n}\right)^{\frac{\alpha-1}{1-\alpha}}}$$

$\Rightarrow$

$$\left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}}(1+n) = \frac{(1-\alpha)\left(\frac{\alpha}{1+n}\right)^{\frac{\alpha}{1-\alpha}}}{2} - \tau$$

$\Rightarrow$

$$\begin{aligned} \tau^* &= \left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}} \left[\frac{(1-\alpha)}{2}\left(\frac{\alpha}{1+n}\right)^{-1} - (1+n)\right] \\ &= \left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}} \left[\frac{(1-\alpha)}{2}\frac{1+n}{\alpha} - (1+n)\right] \\ &= (1+n)\left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}} \left[\frac{(1-\alpha)}{2\alpha} - 1\right] \\ &= (1+n)\left(\frac{\alpha}{1+n}\right)^{\frac{1}{1-\alpha}} \left[\frac{1-3\alpha}{2\alpha}\right] \end{aligned}$$

and $\tau^* > 0$ iff $1 - 3\alpha > 0$ which implies that

$$\frac{1}{3} > \alpha$$

Note that when $\tau = 0$ (7) becomes:

$$k_{t+1}(1+n) = \frac{(1-\alpha)k_t^{\alpha}}{2} \quad (7')$$

and the corresponding steady-state level of capital is

$$k = \left[\frac{(1-\alpha)}{2(1+n)}\right]^{\frac{1}{1-\alpha}}$$

and marginal product of capital is

$$\alpha k^{\alpha-1} = \alpha\left[\frac{(1-\alpha)}{2(1+n)}\right]^{-1} = \frac{2(1+n)\alpha}{(1-\alpha)}$$

The marginal product of capital at the social planner's optimum is:

$$1+n$$

3. **(5 points) Explain your results.**
The problem here is that there are no markets for intergenerational trade, but the government can accomplish intergenerational trade through the tax system. The appropriate lump sum transfers between young and old can achieve a socially optimal steady state. However, the optimal transfers could either flow from the young to the old or from the old to the young depending on whether the competitive equilibrium without taxes has a level of capital above the socially optimal, which is the golden rule level of capital, or below. From the above analysis it seems that the transfer is positive when the share of labor in ouptut, $1-\alpha$, is higher and the share of capital is lower. The reason might be that in those cases young agents have a higher labor income but a low return on capital investment and the only way to attain higher consumption levels upon retirement is to save large amounts of capital, leading to over-acumulation of capital.